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Use of Nintendo Wii Fit™ in the rehabilitation of outpatients following total knee replacement: a preliminary randomised controlled trial☆

Vera Fung, Aileen Ho, Jennifer Shaffer, Esther Chung, Manuel Gomez *

St. John's Rehab Hospital, Toronto, Ontario, Canada

Abstract

Objectives To determine whether Nintendo Wii Fit™ is an acceptable adjunct to physiotherapy treatment in the rehabilitation of balance, lower extremity movement, strength and function in outpatients following total knee replacement.

Design Preliminary randomised controlled trial.

Setting Outpatient department of a rehabilitation hospital.

Participants Adult outpatients following total knee replacement.

Interventions The study group received a physiotherapy session followed by 15 minutes of Wii Fit gaming activities. The games encouraged lateral and multidirectional weight shifting, and provided visual feedback regarding postural balance. The control group received a physiotherapy session followed by 15 minutes of lower extremity strengthening and balance training exercises.

Main outcome measures Length of outpatient rehabilitation, 2-minute walk test, knee range of motion, timed standing, Activity-specific Balance Confidence Scale, Lower Extremity Functional Scale and Numeric Pain Rating Scale, all measured on admission and every 2 weeks until discharge. A patient satisfaction survey was completed at discharge.

Results Seventeen males (34%) and 33 females (66%) with a mean age of 68 (standard deviation 11) years participated in the study. No significant differences in age, gender, days since surgery or length of outpatient rehabilitation were found between the groups. In addition, there were no significant differences in pain ($P = 0.220$), knee flexion ($P = 0.951$), knee extension ($P = 0.492$), walking speed ($P = 0.855$), timed standing tasks ($P = 0.289$), Lower Extremity Functional Scale ($P = 0.079$), Activity-specific Balance Confidence Scale ($P = 0.523$) or patient satisfaction with therapy services ($P = 0.201$) between the groups.

Conclusions Wii Fit is potentially acceptable as an adjunct to physiotherapy intervention for outpatients following total knee replacement, provided the games chosen challenge balance and postural control, and use the lower extremities. Further research is needed to establish whether video games as a therapy adjunct increase patient motivation and compliance with rehabilitation goals.

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Keywords: Total knee replacement; Outpatient; Randomised controlled trial; Video games

Introduction

In recent years, there has been increasing interest in the role of commercially available video game systems in the rehabilitation of various patient populations, as well as increasing availability of clinical literature describing their use. Such video game systems have been applied as a therapeutic intervention option that is considered to be more fun and interesting than conventional exercise [1–4]. Video game therapy has been applied in the rehabilitation of upper extremity [5], lower extremity [6], neurological [7–10] and burn injury patient populations [11–13]. However, few studies have described the use of video games in the rehabilitation of orthopaedic conditions.

Between 2006 and 2007, 37 943 knee replacement operations were performed in Canada, demonstrating a 140% increase in such procedures compared with 1996 to 1997 [14]. Within this 10-year period, the greatest increase in patients electing to undergo total knee replacement (TKR) was seen in 45 to 54 year olds: 271% in males and 337% in females [14].

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* Corresponding author at: St. John's Rehab Hospital, 285 Cummer Avenue, Toronto, Ontario, Canada M2M2G1. Tel.: +1 416 226 6780x7177; fax: +1 416 226 6234.

E-mail address: mgomez@stjohnsrehab.com (M. Gomez).

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The use of a novel, technology-based intervention in rehabilitation is more likely to be viewed as acceptable and valuable in this age group compared with older populations. In addition, video games can offer ways to simulate sport, recreational and functional activities without the intensity or risk associated with performing the actual activity. The Nintendo Wii™ is a commercially available motion controlled video game system (Nintendo of America, Redmond, WA, USA) [15]. Wii Fit™ includes a balance board similar in concept to a force plate. This balance board measures and interprets the pressure distribution of weight applied by the player, and subsequently provides feedback on exercises performed or games played on the accompanying software. This system is marketed to the general public as a method of achieving fitness goals at home using a video game. However, there is potential for this technology to be beneficial in a physical rehabilitation setting. There is a need for randomised controlled trials to compare the benefits of Wii Fit use with traditional rehabilitation therapy [4]. Wii Fit games encourage lower extremity movement, challenge balance, and require the player to remain in a standing position during play. These activities have the potential to address rehabilitation goals involving recovery of lower extremity function. This preliminary randomised controlled study investigated whether Wii Fit use is an acceptable adjunct to physiotherapy treatment in the rehabilitation of balance, lower extremity movement, strength and function in outpatients following TKR.

Methods

Participants

This study was approved by St. John's Rehab Hospital Research Ethics Board, and informed consent was obtained from all participants. The study was conducted between May 2009 and September 2010 in an outpatient department of a rehabilitation hospital in Toronto, Canada. All outpatients with full lower extremity weight bearing, requiring twice-weekly physiotherapy treatment for TKR rehabilitation were eligible for inclusion in the study. Individuals with active painful osteoarthritis in the contralateral lower extremity and those with uncorrected visual impairments were excluded from the study. In addition, as outcomes were measured using self-administered questionnaires, patients with English language barriers or impaired ability to follow instructions were also excluded. A convenience sampling method was used for this study. The treating clinician informed the study personnel about eligible patients following their initial assessment at the outpatient clinic. An investigator provided eligible patients with an information pack about the study, as well as an opportunity to ask questions about study participation. Patients who agreed to participate were randomised, using the randomising function in Microsoft Excel 2003, to the control group or the study group. The control group received 15 minutes of lower extremity exercise, and the study group received 15 minutes of Wii Fit gaming activity. Randomisation was performed at study enrolment. Both control and study interventions were provided in addition to and following each regularly scheduled 60-minute physiotherapy session, in a separate treatment area. Within the 60-minute physiotherapy session, patients received a combination of active and passive knee stretching, lower extremity strengthening and balance exercises, repetitions and sequencing differing between treating clinicians. The study investigators were not the treating clinicians for any of the participants.

Control intervention

Lower extremity exercises that addressed balance, posture, weight shifting and strengthening were provided bilaterally (Appendix 1, see supplementary online material). Exercises, difficulty levels and resistance levels were selected and progressed according to similar exercises already performed successfully in physiotherapy treatment.

Study intervention

Wii Fit games that require the participant to engage in postural control and balance were selected. A four-point walker (Guardian, Model No. 7755, Longmont, Colorado, United States of America) was placed around the balance board for stability, and used by the participant as needed. All participants started by playing the 'Deep Breathing' and 'Ski Slalom' games, but were progressed to other games once they demonstrated a plateau in scoring.

Game descriptions

Lateral weight shifting

- 'Ski Slalom' – the user is required to lean left and right on the balance board to guide the screen character through a series of gates. Sharp turns necessitate increased pressure on the side towards which the user wishes the character to turn.
- 'Tightrope Walk' – the user marches on the balance board to send the screen character across a tightrope. If the character appears to be at risk of falling off the tightrope, the user needs to lean in the opposite direction to correct the character's position before continuing onwards.
- 'Penguin Slide' – the user is represented by a penguin situated in the centre of an iceberg that tilts to the left and right depending on user pressure on the balance board. As the iceberg tilts, the penguin slides across the surface. The aim is to catch as many fish as possible.

Multidirectional balance

- 'Table Tilt' – the user is required to apply pressure to the balance board so that the platform on the screen tilts in the corresponding direction. The goal of this game is to tilt the platform so that marbles sitting on the surface roll into holes in the platform.

- 'Hula Hoop' – while standing on the balance board, the user quickly rotates the hips in a circular motion to enable the character to spin a hula hoop around the waist as many times as possible.
- 'Balance Bubble' – the user is represented by a character inside a bubble situated on a river. The user leans in various directions to guide the bubble down the river, avoiding contact with the riverbank and other obstacles.

Static and dynamic postural control

- 'Deep Breathing' – while the user stands still on the balance board taking slow deep breaths, a tracking dot is displayed showing the user's centre of gravity (COG) and weight distribution. The user is encouraged to guide the tracking dot into the centre of a given target, thereby achieving optimal postural alignment and even weight distribution between the left and right sides.
- 'Half Moon' – the user is instructed to hold arms and clasp fingers over the head and lean slowly at the waist from left to right while maintaining the COG tracking dot steady within the target circle.
- 'Torso Twist' – the user is instructed to hold arms out to the side and twist slowly from left to right while maintaining the COG tracking dot steady within the target circle.

Outcome measures

The following measures were used to track the progress of the participants: active range of motion of the knee (ROM), 2-minute walk test (2MWT) [16,17], Numeric Pain Rating Scale (NPRS) [18–20], Lower Extremity Functional Scale (LEFS) [21,22], Activity-specific Balance Confidence Scale (ABCS) [18,23], and length of outpatient rehabilitation (LOR). Data from the hospital patient satisfaction survey were collected at discharge. Details about the validity and reliability of the standardised outcome measures can be found in Appendix 2 (see supplementary online material) [16–23].

Data collection

Each participant was assessed on the first study visit and every 2 weeks thereafter until they were discharged from physiotherapy services by their treating clinician. Participants did not receive any intervention related to the study on the assessment and re-assessment days. All assessments were conducted by a study investigator who was blinded to the participant's group assignment. Participants were advised not to reveal their group assignment to the assessor.

Data analysis

Data analyses were conducted using Statistical Package for the Social Sciences Version 17.0 (IBM Corp., New York, USA). Initial assessment data (ROM, 2MWT, NPRS, LEFS, ABCS) and participant demographics (age, gender, days since surgery) were compared between the groups.

Table 1
Demographic information of both groups upon study enrolment.

Variable	Study $n=27$	Control $n=23$	P-value
Age in years (SD)	67.9 (9.5)	68.2 (12.8)	0.950[a]
Males (%)	8 (47)	9 (53)	0.480[b]
Females (%)	19 (58)	14 (42)	0.480[b]
Days since surgery (SD)	37.8 (22.1)	46.5 (43.7)	0.393[a]
Length of outpatient rehabilitation in days (SD)	54.2 (27.2)	53.0 (33.7)	0.892[a]

SD, standard deviation.
[a] Student's t-test.
[b] χ^2-test.

Chi-squared analyses were performed for gender. The percentage change in each variable between admission and discharge was analysed and compared between the two groups. Mean percentage change from initial assessment to discharge was calculated for ROM, 2MWT, NPRS, LEFS and ABCS, and compared between the two groups using Student's t-test. Patient satisfaction was measured using a 17-question survey scored on a five-point scale (excellent, very good, good, fair, poor), and analysed using Chi-squared analysis. t-tests were also conducted to compare the mean LOR and number of days since surgery at study onset between the two groups. Exploratory analysis was conducted to compare overall results between patients who required longer outpatient rehabilitation (>6 weeks) with those who were discharged within the expected timeframe for this population ($\leq$6 weeks) across the entire sample, as well as within the control and study groups. P-values < 0.05 were considered to be significant. Standard deviations (SDs) were not available at the time of study initiation; therefore, post-hoc calculations were performed for power and Cohen's d for effect size.

Results

The recruitment rate was 58% (50/86), and all 50 subjects completed the study (Fig. 1, see supplementary online material). Within this sample, there were 17 (34%) males and 33 (66%) females between the ages of 38 and 81 years, with a mean age of 68 (SD 11) years. There were 27 (54%) participants in the study group and 23 (46%) participants in the control group. Demographically, there was no significant difference in mean age, gender, number of days since surgery or LOR between the two groups (Table 1). There was also no difference in participant status at study onset, as measured by the outcome measures selected for assessment (Table 2). Regarding percentage change from admission to discharge, there was no difference in objective measures such as active knee flexion ($P=0.951$), active knee extension ($P=0.492$) and distance covered in the 2MWT ($P=0.855$) between the groups. Subjectively, there was no difference in changes in pain (NPRS, $P=0.115$), self-perceived balance confidence (ABCS, $P=0.523$) and self-perceived lower extremity function (LEFS, $P=0.079$) between the groups (Table 3).

Table 2
Baseline comparison of both groups at study enrolment.

Outcome measure	Study $n=27$	Control $n=23$	P-value[a]
Active knee flexion (degrees)	92.9 (15.6)	93.1 (17.5)	0.96
Active knee extension (degrees)	174.0 (5.8)	174.0 (6.0)	0.59
Two-minute walk test (m)	87.1 (40.0)	90.4 (43.9)	0.79
Numeric Pain Rating Scale (0 to 10, 10 = most pain)	4.7 (2.2)	4.4 (1.9)	0.60
Activity-specific Balance Confidence Scale (0 to 100, 100 = complete confidence)	63.2 (25.2)	64.6 (23.0)	0.83
Lower Extremity Functional Scale (0 to 80, 80 = highest function)	31.8 (15.2)	35.4 (14.4)	0.40

Data shown as mean (standard deviation).
[a] Student's *t*-test.

No difference was found between participants requiring a longer LOR (>6 weeks) compared with those discharged within the expected timeframe (≤6 weeks) (Table 4, see supplementary online material). Post-hoc power calculations for each outcome are shown in Table 5 (see supplementary online material). Cohen's *d* for LEFS was 0.50, indicating a medium effect size and the potential for a large effect size with a sample of 136 participants (68 per group). Patient satisfaction with therapy services did not differ between the two groups ($P=0.201$).

Discussion

The mean age of the study participants was the same as the mean age for the Canadian population that underwent TKR surgery from 2006 to 2007 (68 years) [14]. Also reflective of the Canadian TKR population were the percentage of males (34%) and females (66%) in this study compared with the overall percentage of males (39%) and females (61%) who underwent TKR in 2006 to 2007 [14]. Assuming that such statistics are also reflective for the TKR population of 2009 to 2010, it is expected that the outcomes of this study can be generalised to the cohort of Canadians who required this procedure at the time of this study.

Table 3
Percentage change from study enrolment to discharge between study and control groups.

Outcome Measure	Study $n=27$	Control $n=23$	P-value[*]
Active Knee Flexion	17.18%	17.51%	0.951
Active Knee Extension	0.55%	1.15%	0.492
2 Minute Walk Test	44.29%	41.61%	0.855
NPRS	32.51%	15.67%	0.115
ABCS	48.24%	34.13%	0.523
LEFS	73.36%	41.51%	0.079

NPRS = Numeric Pain Rating Scale.
ABCS = Activity-specific Balance Confidence Scale.
LEFS = Lower Extremity Functional Scale.
[*] Student's *t*-test.

The prevalent use of commercially available video games in the rehabilitation setting warrants the need for stronger research to investigate the efficacy of these systems in providing patient benefit. The objective of this study was to provide information to help guide clinical decision making regarding the use of video games in rehabilitation; an aspect of physiotherapy and occupational therapy practice that continues to grow in acceptance [5,7,8]. Wii Fit appears to be an appropriate intervention tool to address rehabilitation concerns such as balance and standing posture [3,24]. According to a study by Clark *et al*. [25], which was published following completion of data collection for this study, Wii Fit can be considered a reliable and valid measurement tool for standing balance. These findings are congruent with the aim of this study to provide evidence supporting use of Wii Fit as an adjunct intervention for physiotherapy for the rehabilitation of lower extremity conditions. This randomised controlled trial found that the overall outcomes of TKR outpatients who undertook Wii Fit activities did not differ from those of TKR outpatients who undertook physiotherapy exercises, indicating that Wii Fit can be applied as an adjunct intervention without risk of poorer outcomes compared with traditional physiotherapy exercises.

Williams *et al*. [4] conducted a randomised controlled trial to investigate the acceptability of Wii Fit in community-dwelling fallers. Participants were recruited from the community. Williams *et al*. reported a high dropout rate within the group that did not undertake Wii Fit activities, which can be interpreted as an indicator of the ability of video games to increase interest and motivation to exercise. A similar response to randomisation into the conventional exercise group was not experienced in the present study.

The patient satisfaction survey indicated that both the study and control groups enjoyed participating in this study. However, some participants in the study group purposefully expressed enjoyment using Wii Fit, such as 'This is fun' and 'I'm glad I got to be in this (video game) group'. Similar statements were not noted within the control group during participation in study activities. These observations follow the perception that video games provide an interesting and fun treatment option for rehabilitation that has the potential to increase motivation to participate in therapeutic exercise [2,7].

Using a widely available video game or video game application in a study investigating the acceptability of its use as a rehabilitation tool can also be valuable in guiding decision making related to its potential as a home exercise intervention. Although the results from a study conducted in 2007 exploring occupational therapy and physiotherapy perspectives about use of the Wii in rehabilitation did not support home use [26], this could be attributed to clinicians' lack of familiarity with a relatively new technology when the study was conducted. Current literature consists of an increased number of clinical reports and preliminary studies investigating use of the Wii and Wii Fit, as well as its role as a component of home exercise. Both clinical and non-clinical perspectives increasingly support its use to improve balance

and to increase exercise activity at home in various community populations compared with what may have been the case in 2007 [3,4], shortly after the debut of Nintendo Wii in November 2006 [15]. Increased acceptance of the therapeutic potential of Wii Fit, along with the findings of this randomised controlled trial, further support its use as a rehabilitation application. Two of the main outcomes assessed in this study were the LEFS and ABCS. These self-report questionnaires evaluate each participant's perception of how well they are able to perform home and community functions, such as picking up objects from the floor, reaching for objects overhead and walking around the house. If Wii Fit training was introduced as a home exercise or a therapy complement, there might be the potential for transfer of motor movements from the video game exercises to community tasks requiring similar movement abilities. It can then be conceived that ratings on the LEFS and ABCS might improve with regular exercise on the Wii Fit, as would be expected with regular exercise that targets balance and lower extremity function.

It should also be noted that the most common reason why individuals declined to participate in this study was a lack of time to remain in hospital to participate in the study activities. Many of these patients were dependent on others for transportation to and from their therapy sessions, and the flexibility in their schedule was determined by those who were providing transport. For these patients and others in a similar situation, home exercises would be essential to progression of their postoperative recovery. Rationales for the research of video games in rehabilitation often focus on investigating the efficacy of a novel exercise intervention, and its ability to provide activity options that are considered less tedious and more interesting than traditional exercise [2,8,9]. The availability and portability of the Wii also support its implementation as a home exercise option [10]. Applications of Wii Fit as a home exercise intervention aimed at improving basic lower extremity function can be a consideration for clinicians and researchers to explore further.

Study limitations

Based on the frequency of outpatient TKR admission and the targeted time for study completion, this study aimed to recruit 60 participants. However, following the successful recruitment of 50 participants, there was a change in the model of physiotherapy care offered to TKR outpatients at the study hospital. This change may or may not have affected the integrity of the results, but the investigators decided to cease recruitment at this point and complete the study using the data sets collected from 50 participants. In addition, the patient satisfaction questionnaire chosen for administration in this study was the standard survey used across the outpatient department. The objective of this questionnaire is to capture satisfaction related to the delivery of outpatient care, clinician knowledge, and overall quality of hospital services. There was no opportunity to discriminate between responses that reflected perspectives regarding physiotherapy care and experiences of study participation. This was true for both groups. Differences in participant experience between the control and study groups would likely have been better captured using a pre-existing enjoyment scale or one created and piloted specifically for this study. Administration of a different survey following study completion would not have accurately depicted opinions of study participation, as the interval between discharge and the identification of this limitation differed between participants. Due to the variability of activity participation as well as the inability to control for comparable video game exposure of each participant after completion of the study, follow-up was not conducted.

It should be mentioned that despite attempts to maintain exclusivity in the activities performed by the control and study groups in this study, the issue of non-study care exists, as described by Freedland *et al*. [27]. They explained that in randomised controlled trials involving behavioural interventions, the availability of the study intervention from other sources, and the ability of the patients belonging to a control group to access them, makes it difficult to ensure that the effects of the study intervention are completely absent in all control group participants. Although consent packages and study orientation included the request that all participants refrain from all Wii Fit activity, outside of that requested by investigators, for the duration of the study, it was not possible to enforce this request as many participants had access to this equipment outside of the rehabilitation setting. A future study could benefit from the use of newly released video game console peripherals or games, decreasing the likelihood that participants will have access to them during the period of study involvement.

Finally, according to the power calculations conducted for this study (Table 3), the sample size of 50 did not provide sufficient power for the majority of the outcomes assessed. Client-centred care may dictate that subjective- and client-based outcomes such as pain, LEFS and ABCS might be valuable focal measures when assessing the acceptability of a novel intervention in rehabilitation. The power and effect size of the LEFS for this study indicated that Wii Fit had a moderate effect size on self-perceived lower extremity function. Repeating this study with equal-sized study and control groups, each consisting of at least 68 participants, would generate results with more power and would be more indicative of the effect of Wii Fit on LEFS.

Conclusions

This preliminary randomised controlled trial suggests that Wii Fit has potential for use as an adjunct to physiotherapy treatment in the rehabilitation of balance and lower extremity function in outpatients following TKR. Additional studies with larger samples will help to determine the extent to which use of Wii Fit as a physiotherapy adjunct contributes to an improvement in LEFS scores. Further research is required

to establish the role of Wii Fit as a home therapy tool for this population, and whether or not patients receiving therapy via video gaming report greater enjoyment throughout their outpatient admission. The prevalent use of video games in the rehabilitation setting is an indication that they can be motivating and fun options to provide therapeutic activity. There is a need for ongoing research with various patient populations to support this practice in physical rehabilitation.

Ethical approval: St. John's Rehab Hospital Research Ethics Board (REB #2009-02).

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Conflict of interest: None declared.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.physio.2012.04.001.

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